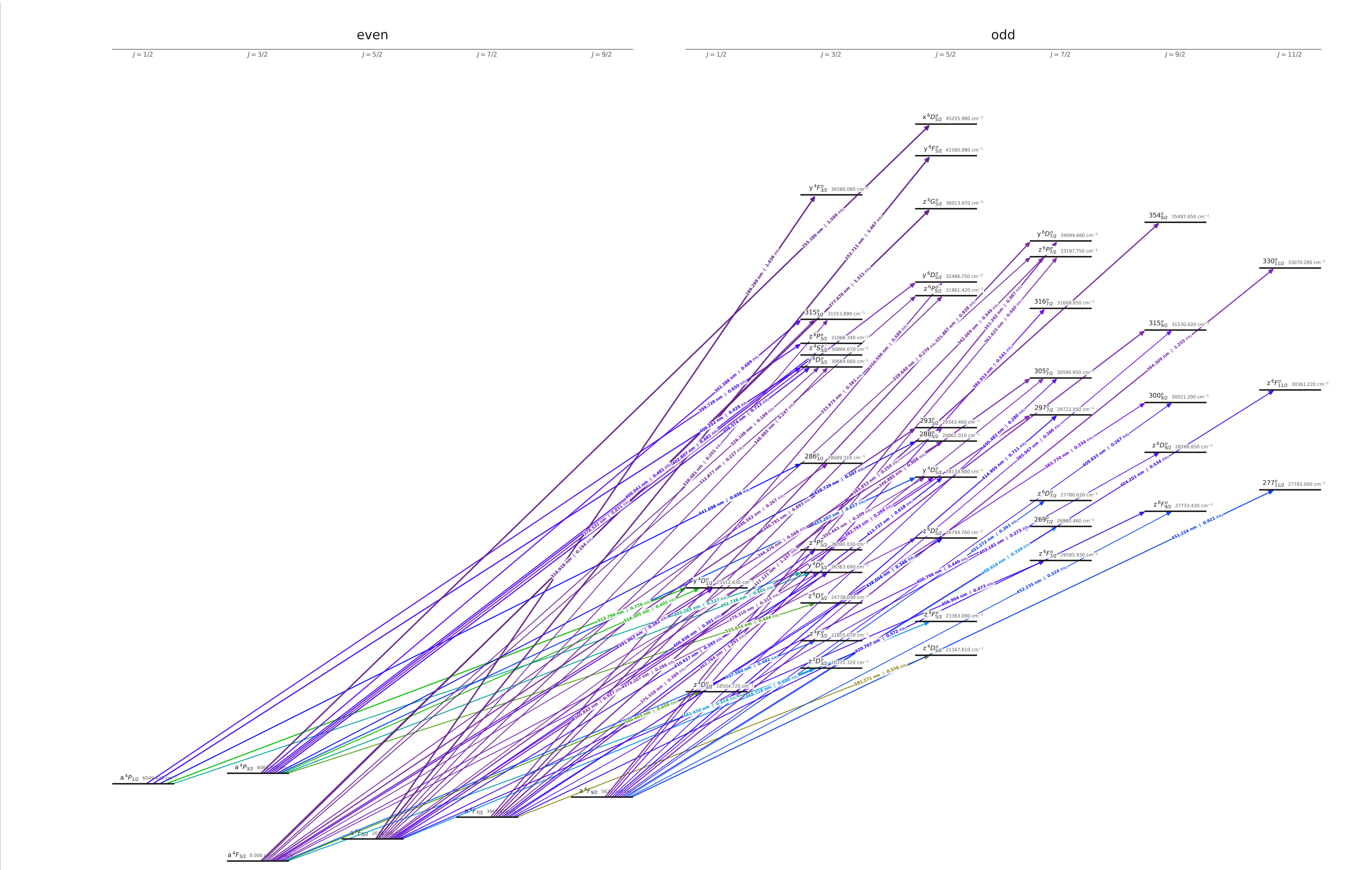


Tantalum

Ta (Ta I). Strongest classified transitions below 2 μm (NIST): 46 levels, 70 lines.

Ta⁺ ionisation limit 60891.40 cm⁻¹ = 7.54957 eV (164.227 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J' \rangle $ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
a ⁴ F _{5/2} - y ⁶ F _{5/2}	252.7111	252.6351	1186.3051	1.467	4.500e+07	-	-	-
a ⁴ P _{3/2} - x ⁶ D _{3/2}	255.1862	255.1097	1174.7988	1.569	5.000e+07	-	-	-
a ⁴ F _{3/2} - z ⁴ G _{5/2}	277.6700	277.5881	1079.6717	1.511	3.600e+07	-	-	-
a ⁴ F _{5/2} - y ⁴ F _{3/2}	289.2685	289.1837	1036.3813	1.636	5.600e+07	-	-	-
a ⁴ F _{7/2} - 354 _{9/2}	317.1207	317.0290	945.3574	1.157	8.500e+06	-	-	-
a ⁴ F _{7/2} - y ⁶ D _{7/2}	331.8870	331.7915	903.2969	0.938	6.100e+06	-	-	-
a ⁴ F _{3/2} - 293 _{3/2}	340.7914	340.6937	879.6948	0.893	6.800e+06	-	-	-
a ⁴ F _{5/2} - 305 _{9/2}	349.8846	349.7845	856.8323	0.908	4.880e+06	-	-	-
a ⁴ F _{9/2} - y ⁴ D _{7/2}	351.2023	351.1018	853.6177	0.987	5.700e+06	-	-	-
a ⁴ F _{5/2} - 297 _{9/2}	360.8434	360.7405	830.8103	0.927	4.630e+06	-	-	-
a ⁴ F _{7/2} - 315 _{9/2}	362.7644	362.6610	826.4109	1.293	7.100e+06	-	-	-
a ⁴ F _{9/2} - 330 _{11/2}	364.3088	364.2050	822.9075	1.255	5.500e+06	-	-	-
a ⁴ P _{3/2} - y ⁶ D _{5/2}	378.5321	378.4246	791.9869	0.831	4.300e+06	-	-	-
a ⁴ P _{3/2} - 315 _{5/2}	392.3880	392.2769	764.0205	0.689	3.980e+06	-	-	-
a ⁴ P _{1/2} - y ⁶ D _{3/2}	406.2524	406.1377	737.9463	0.928	6.500e+06	-	-	-
a ⁴ P _{1/2} - y ⁴ D _{1/2}	513.7899	513.6467	583.4924	0.776	4.500e+06	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ_u).